Samuele Caldarini

Monza (MB), Italy
+39 331-8506706 samuelecaldarini@gmail.com linkedin.com/in/samuele-caldarini-086304232

Experience

XIBER Energy Solutions **Jan. 2026 – Present day**
Data Scientist *Milan, Italy*
- Forecasting, Energy commodities, Simulation, Optimization, AI — Tools: R, Python, SQL, Github

AlgoVerde.ai *Startup* **Dec. 2024 – Dec. 2025**
Data Scientist - Lead [full remote] *San Francisco, California, USA*
- AI agents — Tools: LangGraph (Python), LangSmith, Docker, Github

Brembo N.V. **June 2024 – Nov. 2024**
Data science Intern *Stezzano (BG), Italy*
- Statistical modeling, LLMs, Text data, GenAI — Tools: Azure Databricks (Python, R, SQL), MLflow, Spotfire
- Worked also on my Master's thesis: *Synthetic data and GenAI for improved text classification: a Brembo case study*

Education

Politecnico di Milano **Sep. 2022 – Dec. 2024**
Master of Science in Data Science - Mathematical Engineering *Milan, Italy*

Politecnico di Milano **Sep. 2019 – Sep. 2022**
Bachelor of Science in Mathematical Engineering *Milan, Italy*

Relevant Coursework

- Applied statistics
- Bayesian statistics
- Nonparametric statistics
- Machine learning
- Numerical analysis for Deep learning
- Scientific computing tools for advanced mathematical modeling
- Econometrics
- Networks' complexity
- Game theory

Projects

Surrogate Bayesian Physics-Informed Neural Network for Surface Reconstruction | *Python* **Apr. 2024**
- University project (Pytorch and Git)
- The neural network enables surrogate modeling to provide a quick but accurate estimate of the surface on the entire domain given some sparse data points (Median MSE = 4%)

Extension to the Functional Domain of ITA18 Ground Motion Model | *R, Matlab* **Feb. 2024**
- University project `GMM repo`
- Enhanced the SOTA discrete-time model used by Italian seismologists by improving functional regression techniques. Focus on bootstrap confidence intervals for the continuous-time model's covariates, providing valuable insights for seismologists and enhancing building's safety

Spatio-Temporally Based Deep Survival net for Zillow | *Python, R* **June 2025**
- Personal project (Pytorch)
- By borrowing information from nearby locations and time frames, this framework captures smooth transitions and spatial-temporal heterogeneity. This leads to significantly improved survival predictions compared to one-size-fits-all models or isolated regional analyses.

Skills

Languages: Italian (native), English (C1), French and Spanish (B1)
Coding: Python, R, SQL, Git, Matlab, C++, Latex. Familiar with: Office suite, C, AMPL.
Tools: Positron, VS Code, RStudio, Databricks, GitHub, Posit Connect Cloud, Cloudflare, Shiny, Claude, Gemini, Spotfire, Streamlit, Docker
Other: Data pre-processing, Feature extraction, Geostatistics/Time series/Functional data analysis. Storytelling, Big data, Dashboard maintenance. Attended extra courses about: Probabilistic AI, Causal inference, Deep reinforcement learning, Time series analysis.

Additional info

Further education: Kaplan Int. schools - Dublin. IELTS C1 English language certification
Volunteering: Blood donor (2022 - present). AIM member and mentor (2020 - present)
Interests: swimming, cars, photography, traveling and board games. Favorite movie: Inception
Extra: Merit-based scholarship (BCC Carate & Treviglio) - December 2022